

CHAPTER 1

1.1 Introduction

Pet care has become an integral part of our daily lives, especially for individuals with busy schedules. Feeding pets on time is crucial for their health and well-being, but it can be challenging for pet owners who are away from home. To address this issue, the development of an automatic pet feeder using modern technology provides a practical solution.

This project introduces an Automatic Pet Feeder system that leverages the capabilities of the ESP8266 Wi-Fi module, servo motor, buzzer, and Blynk application to deliver an efficient, user-friendly, and remotely controlled pet feeding experience. The feeder automates the process of dispensing food to pets, ensuring timely feeding while allowing the owner to monitor and control the feeding process via a smartphone.

The ESP8266 acts as the central controller, enabling wireless communication with the Blynk app for remote operation. A servo motor is used to control the food dispensing mechanism, ensuring precise portion control. A buzzer serves as an alert system, notifying the pet or the owner of feeding times. The system is powered by a reliable power supply to ensure continuous operation.

By integrating these components, the automatic pet feeder enhances convenience, reduces the likelihood of overfeeding or underfeeding, and provides a practical solution for pet owners who wish to ensure their pets are well-fed even in their absence. This report outlines the design, implementation, and functionality of the automatic pet feeder, emphasizing its practicality and ease of use.

Pets play a vital role in our lives, offering companionship, emotional support, and reducing stress levels. They help combat loneliness and promote social interaction while contributing to the emotional development and self-esteem of children. However, maintaining a proper feeding schedule for pets can be challenging, especially when pet owners are away or busy. This is where technology, specifically the Internet of Things (IoT), comes into play. IoT can connect devices over the internet, improving quality of life by automating daily tasks, including pet care. With IoT, it becomes possible to automate pet feeding, ensuring pets receive their meals on time without the constant presence of their owners.

One such innovation in IoT is the integration of the ESP8266 microcontroller into pet feeders. The ESP8266 enables remote control via WiFi, managing feeding schedules, processing sensor data, and automating food dispensing. Essentially, it transforms a standard feeder into

a smart device that can be operated from anywhere. The system ensures precise portions are dispensed at regular intervals, promoting better health for pets and reducing food waste. This smart feeder not only provides convenience to pet owners but also guarantees that pets are fed on time and in the correct portions, even in the owner's absence. By harnessing the power of IoT, pet owners can maintain their pets' well-being while also optimizing their own schedules and reducing environmental impact.

Pet owners often face challenges in maintaining consistent feeding schedules due to busy lifestyles or being away from home, which can impact their pets' health. To address this, an automated pet feeder using the ESP8266 microcontroller is needed to ensure timely feeding, accurate portion control, and remote monitoring. This solution would provide convenience for owners and ensure pets are properly cared for, even in the owner's absence.

1.2 Problem Description

Pet care, particularly maintaining a consistent feeding schedule, can be challenging for pet owners with busy lifestyles or unpredictable routines. Missing feeding times or providing inconsistent food portions can negatively impact the health and behavior of pets, leading to issues such as overeating, malnutrition, or stress.

Traditional pet feeding methods require owners to be physically present, which may not always be feasible. While some mechanical feeders exist, they often lack the flexibility of remote control or customization to meet specific feeding requirements. Furthermore, these solutions may not provide real-time feedback or notifications to the owner, limiting their effectiveness in ensuring a pet's well-being.

The primary issues faced by pet owners include:

Inconsistent Feeding Schedule: Pets may not receive meals on time due to the owner's absence or busy schedule.

Overfeeding or Underfeeding: Manual portion control can lead to errors, potentially causing health problems like obesity or malnutrition.

Lack of Monitoring: Owners often lack real-time information about whether their pets have been fed, which can lead to uncertainty and stress.

Inconvenience for Owners: Frequent travel, work commitments, or unexpected delays make it difficult for owners to ensure consistent feeding.

To address these problems, this project proposes an Automatic Pet Feeder system that combines automation, remote control, and real-time notifications.

The system will Ensure timely feeding with accurate portions using a servo motor.

Allow remote control and scheduling via the Blynk application, enabling owners to operate the feeder from anywhere.

Provide alerts through a buzzer to notify the pet of feeding time.

Enhance convenience and reliability through a robust power supply and seamless integration with a smartphone app.

This innovative solution eliminates the stress of managing pet feeding schedules and ensures the pet's health and happiness while providing peace of mind for the owner.

1.3 Motivation

Pet ownership brings joy and companionship, but it also comes with significant responsibilities, particularly in ensuring that pets are well-fed and cared for. Feeding pets on time with the correct portions is essential for their health and well-being. However, maintaining a consistent feeding schedule can be challenging for pet owners with busy work lives, frequent travel, or unpredictable routines. This often leads to irregular feeding patterns, which can negatively impact a pet's physical health and emotional behavior.

The motivation behind developing an Automatic Pet Feeder stems from the following factors:

Pet Health and Well-being: Proper nutrition and timely feeding are crucial for a pet's overall health. Irregular feeding times or incorrect food portions can lead to health issues such as obesity, malnutrition, or behavioral stress. Automating the feeding process ensures consistency, promoting a healthy lifestyle for pets.

Convenience for Owners: Modern lifestyles often demand flexibility and time efficiency. An automatic feeder controlled via a smartphone app eliminates the need for an owner to be physically present during feeding times, providing convenience and peace of mind.

Technological Advancement: With advancements in IoT (Internet of Things) technology, devices like the ESP8266 Wi-Fi module enable smart solutions for everyday problems. Leveraging such technology allows for innovative and user-friendly pet care solutions.

Customized Feeding: Pets have unique dietary requirements based on their size, age, and health conditions. Manual feeding often leads to inconsistent portion control. This system ensures accurate food dispensing, tailored to the specific needs of the pet.

Remote Monitoring and Control: The Blynk application integration enables real-time remote control and monitoring of the feeding process. Owners can easily manage feeding schedules, portion sizes, and receive notifications, even while away from home.

Affordability and Accessibility: While commercial automated feeders exist, many are expensive and lack customization options. This project aims to create an affordable, DIY solution that is easy to build and use, making it accessible to a broader audience.

1.4 Objectives

The objective of your project is to design and implement an automatic pet feeder system using the ESP8266 microcontroller, a servo motor, a buzzer, and the Blynk application for remote control. The system aims to improve the convenience of pet care by automating the feeding process and allowing users to schedule or manually dispense food remotely.

- ❖ Automation of Pet Feeding

To develop an automated system for dispensing food to pets at pre-defined intervals or on demand, reducing the dependency on the pet owner's physical presence.

- ❖ Remote Monitoring and Control

Utilize the Blynk IoT application to enable pet owners to control the feeder remotely via a smartphone, including scheduling feeding times or triggering food dispensing in real-time.

- ❖ Efficient Food Dispensing Mechanism

Implement a servo motor to control the dispensing of food, ensuring precise and consistent portions, reducing wastage, and maintaining a healthy feeding routine for the pet.

- ❖ Notification and Alerts

Use a buzzer to notify the user or pet about feeding times or when food is dispensed, enhancing usability and engagement.

- ❖ Cost-Effective and Reliable Solution

Design the system using low-cost components such as the ESP8266 microcontroller, servo motor, and other readily available parts to make it affordable and accessible for pet owners.

- ❖ Scalability and Customization

Ensure that the system is scalable and adaptable, allowing for modifications to cater to different pet sizes, feeding requirements, and user preferences.

- ❖ Power Efficiency and Backup

Design the system to operate efficiently with minimal power consumption, and consider options for battery backup to ensure reliability during power outages

CHAPTER 2

2.1 Literature Review

The integration of the Internet of Things (IoT) in pet management systems has garnered attention in recent research. This literature survey explores various studies focused on automated feeding solutions, pet health monitoring, and overall pet management through IoT technology.

1. Embedded Interaction with IoT: In their work, Kranz, Holleis, and Schmidt [1] discuss the concept of embedded interaction in the context of the Internet of Things. They explore how smart devices can improve user experience through intelligent interaction, which is vital for developing user-friendly automated pet feeders. The paper highlights the potential of IoT in enhancing daily routines, including pet care, by automating mundane tasks.

2. IoT Research and Applications in Pet Systems: A study published in 2019 examines various IoT applications in pet systems [2]. This research outlines innovative solutions for monitoring pet health, safety, and feeding habits, demonstrating the effectiveness of IoT in enhancing pet management. It emphasizes the importance of continuous research and development in creating robust pet care systems that adapt to owner needs and improve pet well-being.

3. Review of IoT in Pet Management:

Jigar Mashekar and Sharma [3] provide a comprehensive review of IoT applications in pet management. Their analysis includes various systems designed for tracking pet activities, health monitoring, and automated feeding. The authors discuss key performance indicators (KPIs) that influence the effectiveness of these systems, such as user satisfaction, system reliability, and cost-effectiveness.

4. IoT in Home Automation

The Internet of Things (IoT) has transformed traditional systems into intelligent and connected solutions. Studies highlight how IoT-enabled devices, like ESP8266 microcontrollers, facilitate remote monitoring and control through mobile applications. Researchers have successfully demonstrated the integration of IoT in home automation systems, including smart lighting, security, and pet care solutions, enhancing convenience and efficiency.

ESP8266 Microcontroller: Widely recognized for its low cost, Wi-Fi capability, and ease of use, the ESP8266 is commonly employed in IoT projects for remote communication with devices via smartphone apps such as Blynk. Its compatibility with various sensors and actuators makes it an ideal choice for IoT-based automatic pet feeders.

5. Automatic Pet Feeders

Automatic pet feeders have been developed to assist pet owners by automating the food dispensing process. Traditional feeders typically rely on mechanical timers or manual settings, but IoT-based feeders introduce advanced functionality, including remote scheduling, real-time control, and notifications.

Servo Motor Mechanisms: Servo motors are commonly used for precise control of food dispensing. Studies reveal that servo motors provide reliable and efficient movement to manage the opening and closing of food containers or dispensing mechanisms.

Challenges: Research indicates challenges such as food clogging, portion control accuracy, and limited remote interaction in earlier designs.

7. Role of Blynk Application in IoT Projects

Blynk is a popular IoT platform that allows users to create customized control panels for smart devices via smartphones. Its integration with the ESP8266 enables real-time communication between the hardware and the user. Studies demonstrate the effectiveness of Blynk in controlling home automation systems, especially for applications requiring timely updates, scheduling, and interactive controls.

Advantages: Blynk's user-friendly interface and support for cloud-based communication simplify the development of IoT systems, making it suitable for pet feeder applications.

8 Food Dispensing Mechanisms and User Interaction

Research on food dispensing mechanisms emphasizes the importance of reliability and accuracy. The integration of buzzers in automated systems has been highlighted in various studies for providing audio feedback to enhance user interaction and alert pets during food dispensing.

Buzzer as a Notification Device: Buzzers serve as an essential component to notify users of actions such as food dispensing or system errors, adding a layer of engagement and functionality.

9. Power Supply and Efficiency in IoT Systems

Efficient power usage is critical for IoT systems, particularly in devices designed for continuous or scheduled operation. Research shows that ESP8266, combined with low-power modes and optimized programming, can significantly extend the system's operational life, even when powered by batteries. Backup power options are often recommended for critical systems like pet feeders to ensure uninterrupted functionality.

10. Similar Projects and Case Studies

Several academic and DIY projects have explored automatic pet feeders. These projects often employ ESP8266, servo motors, and mobile applications for control, providing valuable insights into system design, common pitfalls, and user feedback.

11. Notable Findings:

Remote scheduling greatly improves the convenience of feeding pets.

Real-time notifications via IoT platforms help users track feeding activities.

Integration of cameras or additional sensors enhances usability but increases system complexity and cost.

STUDY	FOCUS AREA	KEY FINDINGS	KPIs
Kranz et al,[1]	Embedded Interaction	Emphasizes user-friendly interfaces for pet care.	User satisfaction,ease of use.
IOT research[2]	Health &safety Monitoring	Demonstrates effectiveness in monitoring pet well being.	Reliability adaptibility
Jigar Mashekar&Sharma[3]	Comprehensive Iot application	Analyzes various pet management system	Cost effectiveness,user engagement

CHAPTER 3

3.1 Methodology:-

3.1.1 Block Diagram

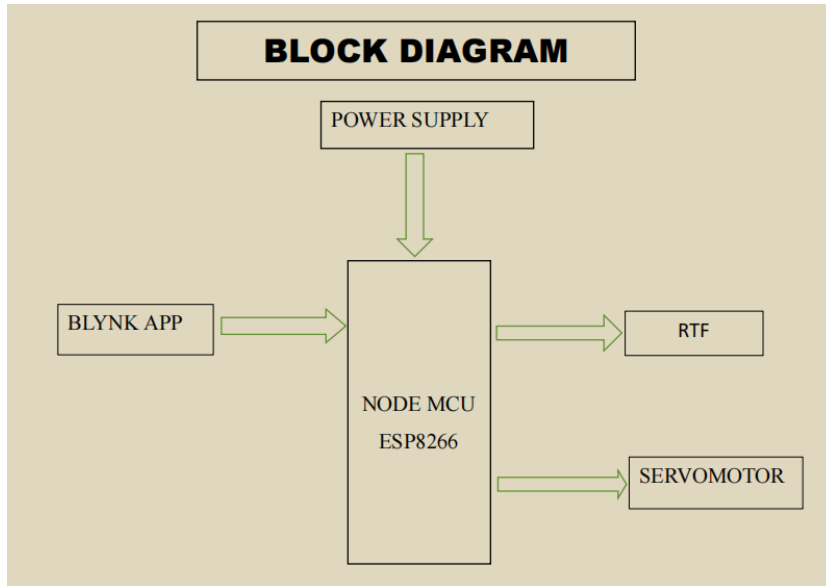


Fig 3.1 :Block Diagram of Automatic pet Feeder

The development of the automated pet feeder using the ESP8266 module follows a systematic approach to ensure the proper functionality of the system. The steps below outline the key stages involved, including component selection, circuit design, programming, testing, and final assembly. These steps are essential to achieving the project's objectives.

a. Components:

The following components will be used to build the automated pet feeder:

- 1) **ESP8266:** The main microcontroller for WiFi connectivity and system control.
- 2) **Servo Motor:** Used to control the dispensing mechanism for food.
- 3) **Power Supply:** A 3.3V supply for the ESP8266 and a 5V supply for the servo motor.
- 4) **Buzzer:** Provides alerts for feeding events or errors.
- 5) **Wires and Breadboard/PCB:** For connecting the components together.
- 6) **Food Container:** Holds the pet food, which will be dispensed by the system.

b. Circuit Design:

The circuit will be designed to integrate the servo motor, buzzer, and ESP8266 in a stable configuration.

- 1) **Servo Motor:** Connect the signal pin of the servo motor to the D3 pin on the ESP8266.
- 2) **Buzzer:** Connect the buzzer to the D4 pin of the ESP8266 for audio alerts.

The development of the automated pet feeder using the ESP8266 module follows a systematic approach to ensure the proper functionality of the system. The steps below outline the key stages involved, including component selection, circuit design, programming, testing, and final assembly. These steps are essential to achieving the project's objectives.

3) Software tools:

- a. Arduino IDE(for programming ESP8266)
- b. Blynk Application (for IoT control)

c. Programming:

Using the Arduino IDE, the ESP8266 will be programmed to control the feeder.

- 1) Include the 'Servo.h' library to control the servo motor for food dispensing.
- 2) Code the ESP8266 to trigger the servo motor at scheduled times to dispense food.
- 3) Program the buzzer to sound alerts when food is dispensed or if an error occurs.
- 4) Create a mobile/web interface to allow remote control and monitoring.

d. Testing:

- 1) Servo Motor Calibration: Calibrate the servo to ensure it dispenses the correct amount of food each time. Adjust the angle and speed of rotation to match the dispensing mechanism.
- 2) Buzzer: Test the buzzer for proper alert functioning during feeding or when the system encounters issues.
- 3) System Integration: Test the overall system to ensure all components (servo, buzzer, ESP8266) work together smoothly and remotely via the web interface.

e. Assembly:

Attach the servo motor to the food dispensing mechanism (such as a rotating door or flap) that

releases food from the container. Place all components (ESP8266, servo motor, power supply, buzzer) securely in an enclosure to protect the system and ensure stability during operation. Ensure that the food container is properly aligned with the dispensing mechanism.

1) Hardware Assembly:

- a. **Food Dispenser:** Use a container with a rotating flap mechanism controlled by the servo motor to dispense food in measured quantities.
- b. **Buzzer Setup:** Place the buzzer to provide audible notifications for feeding times or errors.
- c. **Power Supply:** Connect the ESP8266 and servo motor to the appropriate power sources, ensuring voltage and current requirements are met.

2) Software Development:

a. Programming ESP8266:

- i. Use the Arduino IDE to write and upload the program.
- ii. Import necessary libraries:
- iii. Servo.h (for servo control)
- iv. BlynkSimpleEsp8266.h (for Blynk integration)

b. Code functionalities:

- i. Servo motor rotation for dispensing food.
- ii. Buzzer activation for notifications.
- iii. Wi-Fi setup for ESP8266.
- iv. Blynk button and timer functionalities

c. Blynk App Configuration:

- i. Create a project in the Blynk app.

d. Add widgets:

- i. Button: To manually control food dispensing.
- ii. Timer: To schedule feeding times.
- iii. Notification: For reminders or alerts.
- iv. Configure virtual pins in the app

3.1.2 Circuit Diagram

(add picture of circuit diagram)

CHAPTER 4

4.1 Result

The development of the automated pet feeder using the ESP8266 module results in a robust, IoT-based system that automates the feeding process for pets. The system incorporates the ESP8266 microcontroller for WiFi connectivity, a servo motor to dispense food, and a buzzer for user notifications. Below is a detailed explanation of the key outcomes:

1) System Functionality

- a. **Automated Feeding:** The ESP8266 microcontroller controls the servo motor to dispense food from a container at predefined times, ensuring pets are fed automatically without the need for manual intervention. The servo motor's position is adjusted based on the programmed schedule, delivering the correct portion of food.
- b. **Remote Control via Blynk App:** The Blynk app acts as the user interface, allowing pet owners to control the system remotely. Users can schedule feeding times, manually trigger food dispensing, and receive alerts or notifications about the system's status.
- c. **Buzzer Alerts:** A buzzer is integrated into the system to provide audible notifications. This buzzer signals successful feedings, alerts the user about potential errors (e.g., low food levels or system malfunction), and ensures the user is informed of the system's activity even without checking the app.

2) Component Integration

- a. **ESP8266 Module:** The ESP8266 provides the essential WiFi connectivity for the system, enabling remote operation via the Blynk app. The module is programmed using the Arduino IDE to handle food dispensing control and communication with the app.
- b. **Servo Motor:** The servo motor is controlled by the ESP8266 to rotate a flap or door mechanism in the food container. The servo motor is calibrated to dispense a specific amount of food each time, ensuring that pets are fed properly. The rotation angle and speed are fine-tuned for optimal operation.
- c. **Buzzer:** The buzzer is connected to the ESP8266 and used to indicate feeding events and errors. The buzzer ensures that users are notified of important system statuses, such as successful food dispensing or a malfunction.

-
- d. Power Supply:** Two distinct power supplies are used – a 3.3V supply for the ESP8266 module and a 5V supply for the servo motor. This ensures each component operates at its required voltage and performs efficiently.

3) Software Development

- a. The ESP8266 is programmed using the **Arduino IDE**, which involves importing libraries such as Servo.h (for controlling the servo motor) and BlynkSimpleEsp8266.h (for integrating the system with the Blynk app).
- b. The system's functionalities include:
- i. Servo motor control for dispensing food at specific times.
 - ii. Buzzer activation to alert users about successful feedings or errors.
 - iii. WiFi connectivity setup to connect the ESP8266 to the internet, ensuring remote control and monitoring through the Blynk app.
 - iv. Timer and button widgets in the Blynk app to schedule feeding times and manually trigger food dispensing.

4) Testing Results

- a. Servo Motor Calibration:** The servo motor was successfully calibrated to ensure the accurate dispensing of food at each feeding. Adjustments to the angle and rotation speed were made to match the desired portion size.
- b. Buzzer Functionality:** The buzzer operated as intended, providing clear alerts when the food was dispensed or when an error occurred. The alert notifications were audible and easy to recognize, confirming the system's operational status.
- c. System Integration:** All components (ESP8266, servo motor, buzzer) worked seamlessly together. The system was tested both for automatic operation through the scheduled timer and for manual operation through the Blynk app, and both functions performed reliably.

5) System Assembly

- a. Hardware Assembly:** The servo motor was securely attached to the food dispensing mechanism (a rotating flap or door) to control the release of food. All components, including the ESP8266, servo motor, power supply, and buzzer, were enclosed in a protective housing to ensure stability and safeguard the components.
- b. Food Container Setup:** The food container was aligned with the servo mechanism, ensuring smooth food release when the servo motor activated.

- c. **Power Connections:** The system was carefully wired to ensure proper voltage distribution: the ESP8266 was powered by a 3.3V supply, while the servo motor received 5V. This ensured that both components worked optimally without power-related issues.

6) Blynk App Integration

- a. **User Interface Configuration:** The Blynk app provided an intuitive interface for controlling the pet feeder. Widgets like **buttons** for manual food dispensing, **timers** for scheduling feeding times, and **notifications** for system alerts were configured in the app.
- b. **Virtual Pin Mapping:** Virtual pins in the Blynk app were mapped to control the servo motor (for food dispensing), buzzer (for alerts), and other system functionalities. This integration allowed users to remotely monitor and manage the pet feeder through their smartphones or web browser.

7) Final Outcome

The final result is a fully operational automated pet feeder system that:

- a. Dispenses food at scheduled times or on-demand via manual control from the Blynk app.
- b. Alerts the user with a buzzer for successful feedings and error notifications.
- c. Provides remote monitoring and control, making it easy for pet owners to manage their pets' feeding schedule, even when they are away from home.
- d. Operates reliably and efficiently, with accurate portion dispensing and system alerts.

(add picture of result)

CHAPTER 5

5.1 Conclusions and Future Scope

5.1.1 Conclusion

The automated pet feeder system using the ESP8266 module has successfully achieved its goal of automating the pet feeding process. By leveraging IoT technology and the Blynk app, the system enables remote control and scheduling of feeding times, ensuring pets are fed on time without the need for manual intervention. The integration of the ESP8266 microcontroller, servo motor, and buzzer offers reliable food dispensing and user notifications, providing an efficient and user-friendly solution for pet owners. The system's performance during testing, including servo motor calibration, buzzer alerts, and seamless app integration, confirms its functionality and reliability.

This project not only simplifies pet care for owners but also ensures that pets receive consistent feeding, reducing the chances of overfeeding or underfeeding. The modular nature of the system also opens the door for future enhancements and improvements.

5.1.2 Future Scope

While the current automated pet feeder system is functional and efficient, there are several opportunities for expansion and enhancement in the future:

- 1) Integration with Other IoT Devices:** The system can be expanded to integrate with other IoT devices, such as cameras for monitoring pets or sensors to detect food levels in the container, triggering alerts when food is low or the system is about to run out of food.
- 2) Mobile App Enhancements:** Future versions of the Blynk app or a dedicated mobile app can include more advanced features, such as feeding history logs, real-time monitoring of the pet's food intake, and even video streaming to monitor pets during feeding times.
- 3) Machine Learning for Feeding Schedules:** Implementing machine learning algorithms could allow the system to learn and adapt to a pet's feeding behavior, such as adjusting portion sizes or feeding times based on the pet's activity or consumption patterns.

- 4) **Voice Control Integration:** Integration with voice assistants like Amazon Alexa or Google Assistant could allow pet owners to control the feeder through voice commands, providing an added level of convenience and ease of use.
- 5) **Energy Efficiency Improvements:** Further optimization could be made to reduce the system's power consumption, such as implementing power-saving modes when the system is idle, ensuring that the pet feeder is energy-efficient in long-term use.
- 6) **Multiple Pet Support:** The system could be designed to support multiple pets, allowing the user to schedule feeding times and control food portions for different pets independently. This could be achieved by adding more dispensing mechanisms and servo motors for each pet.
- 7) **Automated Food Refilling:** A future version of the feeder could incorporate an automatic food refill system, possibly using sensors to detect when the food level is low and triggering an automated refill mechanism or alerting the user to refill the container.

By incorporating these advancements, the automated pet feeder system can become even more intelligent, efficient, and capable of meeting the evolving needs of pet owners. The project has the potential to be expanded into a comprehensive pet care solution that provides convenience, automation, and greater peace of mind for pet owners.

References

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Appendices

Code:-

```
#define BLYNK_TEMPLATE_ID "TMPL3MQMadnLu"
#define BLYNK_TEMPLATE_NAME "Sai ram"
#define BLYNK_AUTH_TOKEN "BeZ6vXY2Z8tm2rL3uJG5CXIXdXtAF9mD"

#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <Servo.h>

// Blynk authentication token
char auth[] = "BeZ6vXY2Z8tm2rL3uJG5CXIXdXtAF9mD";

// WiFi credentials
char ssid[] = "Druva";
char pass[] = "12345678";

// Pin Definitions
Servo myServo;
const int servoPin = D4; // Change as per your connection
const int buzzerPin = D2; // Connect the buzzer to D2
int sweepFlag = 0; // To track the sweep state
int angle = 0; // Initial servo angle

// Blynk Virtual Pin
#define V0 0 // Switch widget in Blynk app

// Blynk Write Function
BLYNK_WRITE(V0) {
    sweepFlag = param.asInt(); // Read switch state (1 for ON, 0 for OFF)

    // Control the buzzer based on switch state
    if (sweepFlag) {
        digitalWrite(buzzerPin, HIGH); // Turn on buzzer
```

```

    } else {
        digitalWrite(buzzerPin, LOW); // Turn off buzzer
    }
}

// Sweep function
void sweepServo() {
    for (angle = 0; angle <= 180; angle++) {
        if (!sweepFlag) return; // Stop sweeping if switch is OFF
        myServo.write(angle);
        delay(15);
    }
    for (angle = 180; angle >= 0; angle--) {
        if (!sweepFlag) return; // Stop sweeping if switch is OFF
        myServo.write(angle);
        delay(15);
    }
}

void setup() {
    // Initialize serial monitor
    Serial.begin(115200);

    // Setup servo
    myServo.attach(servoPin);
    myServo.write(90); // Set servo to middle position

    // Setup buzzer
    pinMode(buzzerPin, OUTPUT);
    digitalWrite(buzzerPin, LOW); // Ensure buzzer is off initially

    // Initialize Blynk
    Blynk.begin(auth, ssid, pass);
}

```

```

    // Initialize Blynk
    Blynk.begin(auth, ssid, pass);

    // Debug messages
    Serial.println("ESP8266 and Blynk are ready.");
}

void loop() {
    Blynk.run(); // Run Blynk
    if (sweepFlag) {
        sweepServo();
    }
}

```

